

Warehouse Wizard × NetSuite — NetSuite Developer Brief

Sep 24, 2026 · @Russell Hunte

Jan, two items on your side gate the project: uploading the M2M certificate so the connection can authenticate, and a small RESTlet to receive inbound shipments, which NetSuite's REST API cannot do through a transform. This brief covers both, plus the record-level detail for the rest of the integration.

Where the integration stands today

Area	State
Authentication	OAuth 2.0 client credentials (M2M) with an ES256-signed JWT client assertion. Warehouse Wizard generates an EC P-256 key and self-signed certificate.
Inbound: items	Processed. Item records upsert Warehouse Wizard products, linked by NetSuite internal id
Inbound: POs, SOs, transfer orders, fulfilments	A webhook accepts them and parks them in a queue. No processor exists yet; this project builds them
Outbound: inventory adjustment	Built, not yet accepted live. Posts to the inventory adjustment record with an <code>externalId</code> of the form <code>ww-inventory-adjustment-<key></code> for idempotency; after a timeout it checks <code>inventoryAdjustment/eid:<externalId></code> . It needs an adjustment GL account id before it will run
Item lookup	The admin screen runs SuiteQL against <code>item</code> to pick items to import
Location mapping	Warehouse Wizard warehouses map to NetSuite location internal ids in Settings

The same `externalId` pattern will be used for every new record type, so a retry after a timeout never creates a duplicate.

Integration points

All calls go to `/services/rest/record/v1/` unless marked otherwise.

Warehouse action	Call	Phase
Look up a PO, inbound shipment, transfer order, sales order or RMA	GET the record and its line sublist; SuiteQL (<code>/services/rest/query/v1/suiteql</code>) for search and lists	1-3
Receive a purchase order	POST <code>purchaseOrder/{id}/!transform/itemReceipt</code>	1
Receive an inbound shipment	No transform exists — needs the RESTlet described below	1
Receive a transfer order	POST <code>transferOrder/{id}/!transform/itemReceipt</code>	1
Ship a transfer order	POST <code>transferOrder/{id}/!transform/itemFulfillment</code>	2
Ship a sales order	POST <code>salesOrder/{id}/!transform/itemFulfillment</code> , then PATCH <code>shipStatus</code> from Picked to Shipped	2
Receive a customer return	POST <code>returnAuthorization/{id}/!transform/itemReceipt</code>	3
Move stock between bins	POST <code>binTransfer</code>	2-3
Correct a count variance	POST <code>inventoryAdjustment</code> (already built)	1

Every line carries lot, serial, bin and status detail in the `inventoryDetail` subrecord (`inventoryAssignment.items[]`), using `receiptInventoryNumber` on receipts and a reference to an existing inventory number on fulfilments. Line numbers are not sequential, so Warehouse Wizard reads the source record's lines before every transform and records the source line number on its own lines.

Inbound triggers will be user-initiated lookups by document number rather than scheduled polling, in line with your concern about integration capacity. A User Event script pushing to the existing webhook remains an option if near-real-time notification is wanted later.

The inbound shipment receiving service

Why it is needed

The `inboundShipment` REST record supports create, read, update, delete and search, but not transform ([Oracle: Inbound Shipment REST record](#)). Receiving uses the separate `receiveinboundshipment` record, which Oracle documents only for SuiteScript ([Oracle: Inbound Shipment scripting](#)). As Wilson noted, once a PO is on an inbound shipment it can only be received through the shipment, so transforming the PO directly is not a fallback. Question J6 asks you to confirm this in the sandbox.

Proposed contract

A RESTlet (POST), authenticated by the same M2M integration and role.

```
{
  "inboundShipmentId": "123",
  "externalRef": "ww-receipt-RCV-0042",
  "tranDate": "2026-10-05",
  "lines": [
    {
      "shipmentLineKey": "4827",
      "quantity": 95,
      "location": "6",
      "inventoryAssignment": [
        { "receiptInventoryNumber": "LOT-A1", "binNumber": "20",
"inventoryStatus": "1", "expirationDate": "2027-09-01", "quantity": 95 }
      ]
    }
  ]
}
```

```
{ "ok": true, "itemReceiptIds": ["8841", "8842"], "alreadyProcessed": false }
```

Behaviour

1. Load `receiveinboundshipment` by the shipment id (dynamic mode).
2. For each line in the request, select the matching `receiveitems` line, set `quantitytobereceived`, the receiving location, and the inventory detail.
3. Set lines not in the request to zero.
4. Save, then return the item receipt ids NetSuite created (one per PO).

5. **Idempotency:** before loading, search item receipts for `externalRef` (stored in a custom body field or the memo). If found, return those ids with `alreadyProcessed: true` rather than receiving again.
6. On a validation error (quantity above expected, closed shipment, locked posting period) return `ok: false` with NetSuite's message, so Warehouse Wizard can show it to the operator.

The field names above are indicative. Adjust them to whatever the account's record browser shows, and tell us the final shape.

Beta sandbox setup

A full copy of Warehouse Wizard will run against the NetSuite sandbox while production continues unchanged. The beta must be **unable** to reach live NetSuite, and that is enforced by giving it separate credentials.

1. Create a **new integration record in the sandbox** with the Client Credentials (Machine to Machine) grant and the REST Web Services scope. Do not reuse the production record.
2. Russell generates a separate certificate from the beta's Settings → Integrations screen and sends you the public certificate.
3. Upload it under Setup → Integration → Manage Authentication → OAuth 2.0 Client Credentials (M2M) Setup, mapping the integration record, an entity and a role.
4. Send back the sandbox account id, the client id and the Certificate ID NetSuite assigns.
5. Deploy the inbound shipment RESTlet to the sandbox first, and give the same role permission to run it.

Role permissions needed: create and edit on Item Receipt, Item Fulfillment, Bin Transfer and Inventory Adjustment; view on Purchase Order, Inbound Shipment, Transfer Order, Sales Order, Return Authorization, Location, Bin, Item and Inventory Status; REST Web Services; and SuiteAnalytics Workbook for SuiteQL.

Sandbox test data needed before acceptance testing:

- One local PO
- One inbound shipment carrying two POs
- One transfer order between two sandbox locations
- One sales order for a multi-unit item, such as the rice bale
- One return authorisation
- At least one lot-tracked item, and one location with bins

For production, the same steps apply to the live integration record. The certificate is ready to download from Settings → Integrations → Signing certificate.

Things to verify in the sandbox

- **Line identification on transforms.** Oracle's examples use `orderLine` on fulfilments and `line` on receipts. Confirm which the item receipt transform accepts for POs, transfer orders and RMAs.
- **Inbound shipment line key.** `shipmentitem` must be the PO line's `lineuniquekey`, not the item's internal id.
- **Transfer order receipts with several fulfilments.** Can the REST transform name which fulfilment is being received? If not, the same RESTlet approach may be needed there, using `record.transform` with the fulfilment in `defaultValues`.
- **Inventory detail field casing** per record. Take it from `GET /services/rest/record/v1/metadata-catalog/<record>` with `Accept: application/swagger+json`, rather than from documentation.
- **Status codes.** Confirm the letter codes for `PurchOrd`, `TrnfrOrd`, `RtnAuth`, `SalesOrd` and `ItemShip` in this account.
- **Restock field id** on RMA item receipt lines.
- **WMS coexistence.** If NetSuite's warehouse module is active at a location, can a fulfilment be created directly without a wave?

Read-only discovery queries

Once the certificate is in, these answer several open questions without anyone checking screens by hand.

```
-- Locations and bin usage
SELECT id, name, BUILTIN.DF(subsidiary) AS subsidiary, usebins,
makeinventoryavailable, isinactive
FROM location ORDER BY name;

-- Bins per location
SELECT BUILTIN.DF(location) AS location, COUNT(*) AS bins FROM bin GROUP BY
BUILTIN.DF(location);

-- Inventory statuses (fails if the feature is off)
SELECT id, name, inventoryavailable FROM inventorystatus;

-- Lot, serial and bin mix across active items
SELECT itemtype, islotitem, isserialitem, usebins, COUNT(*) AS n
FROM item WHERE isinactive = 'F' GROUP BY itemtype, islotitem, isserialitem,
```

```

usebins;

-- Status codes actually in use
SELECT type, status, BUILTIN.DF(status) AS label, COUNT(*) AS n
FROM transaction WHERE type IN
('PurchOrd', 'TrnfrOrd', 'RtnAuth', 'SalesOrd', 'ItemShip', 'ItemRcpt')
GROUP BY type, status, BUILTIN.DF(status);

-- Inbound shipments in flight (fails if the feature is off)
SELECT id, shipmentnumber, externaldocumentnumber, BUILTIN.DF(shipmentstatus)
AS status
FROM inboundshipment;

-- NetSuite warehouse module footprint (fails if not installed)
SELECT COUNT(*) FROM wave;

```

Feature switches such as Advanced Receiving and Advanced Shipping cannot be read through SuiteQL; Wilson is confirming those.

Your questions and tasks

#	Question	Why it matters
J1	Can the production certificate be uploaded now?	Blocks everything. The connection is built and fails only at authentication
J2	Can you confirm the inbound shipment analysis and build the RESTlet?	Blocks Phase 1 acceptance
J3	Which GL account (and subsidiary, if needed) should inventory adjustments post to?	The adjustment worker will not run until it is set
J4	What is the account's concurrency limit, and how much of it is already in use?	Sets how many requests Warehouse Wizard may run in parallel
J5	Can you set up the sandbox integration record and certificate for the beta?	The beta must not reach live NetSuite
J6	If a PO on an inbound shipment is received directly, does the shipment's received quantity update, or does NetSuite refuse?	Confirms there is no REST fallback
J7	When can the PO number be added to the printed receiving report?	The receiving flow starts from that sheet

#	Question	Why it matters
J8	What did your pallet test find: custom record, or pallet number on item records?	Recommendation is to keep pallets in Warehouse Wizard; your result should be recorded
J9	If NetSuite's warehouse module is active at a location, can a fulfilment be created there without a wave?	Decides whether both systems can run side by side
J10	Can you confirm the field names and line identifiers from the checks above?	Avoids rework during the build

Tasks

- ☐ Upload the production M2M certificate and confirm Test connection passes
- ☐ Create the sandbox integration record and upload the beta certificate
- ☐ Build and deploy the inbound shipment RESTlet to the sandbox (about 2–4 days)
- ☐ Add the PO number to the receiving report
- ☐ Create the mirrored bins for Pine and Lower Estate, with Wilson
- ☐ Set up sandbox test data
- ☐ Run the discovery queries and share the results

The full proposal, including cost, phasing and the complete open-questions list, accompanies this brief.